1. Nine binary integer variables represent each project and region type pairing. The objective is to maximize NPV while adhering to a $15M budget cap, one project for each region, and two projects overall. The optimal solution is a $11M NPV done by building a wind turbine in the south ($5M NPV) and a geothermal plant in the north ($6M NPV). This solution costs $14M, no region shares a project and there are two projects.

2. The program uses seven integer variables for workers on each three-day shift pattern and seven for daily part-time hires at 800 dollars. The objective minimizes total cost, where each shift worker earns their daily wage across all three days, subject to meeting each day's crew requirement. The optimal cost is $297,800, using Shift 1 (100), Shift 2 (100), Shift 4 (70), Shift 5 (180), and Shift 6 (50), with no part-time workers.

3. There are 120 binary variables assigning 15 items to 8 boxes, plus 8 variables marking whether each box is used. The objective minimizes boxes used, subject to each item going in exactly one box and no box exceeding its 50 cubic foot capacity. The optimal solution uses 5 boxes to hold all 215 cubic feet of items, saving 3 boxes. Since 215 cubic feet needs at least 5 boxes of 50.

4. 64 binary variables are assigned 8 shifts to 8 possible guards, plus 8 variables marking whether each guard is hired. The objective minimizes guards hired, subject to each shift being covered once and no two conflicting shifts sharing a guard. The optimal solution hires 3 guards. Since shifts 1, 2, and 3 all mutually conflict, at least 3 guards are required.